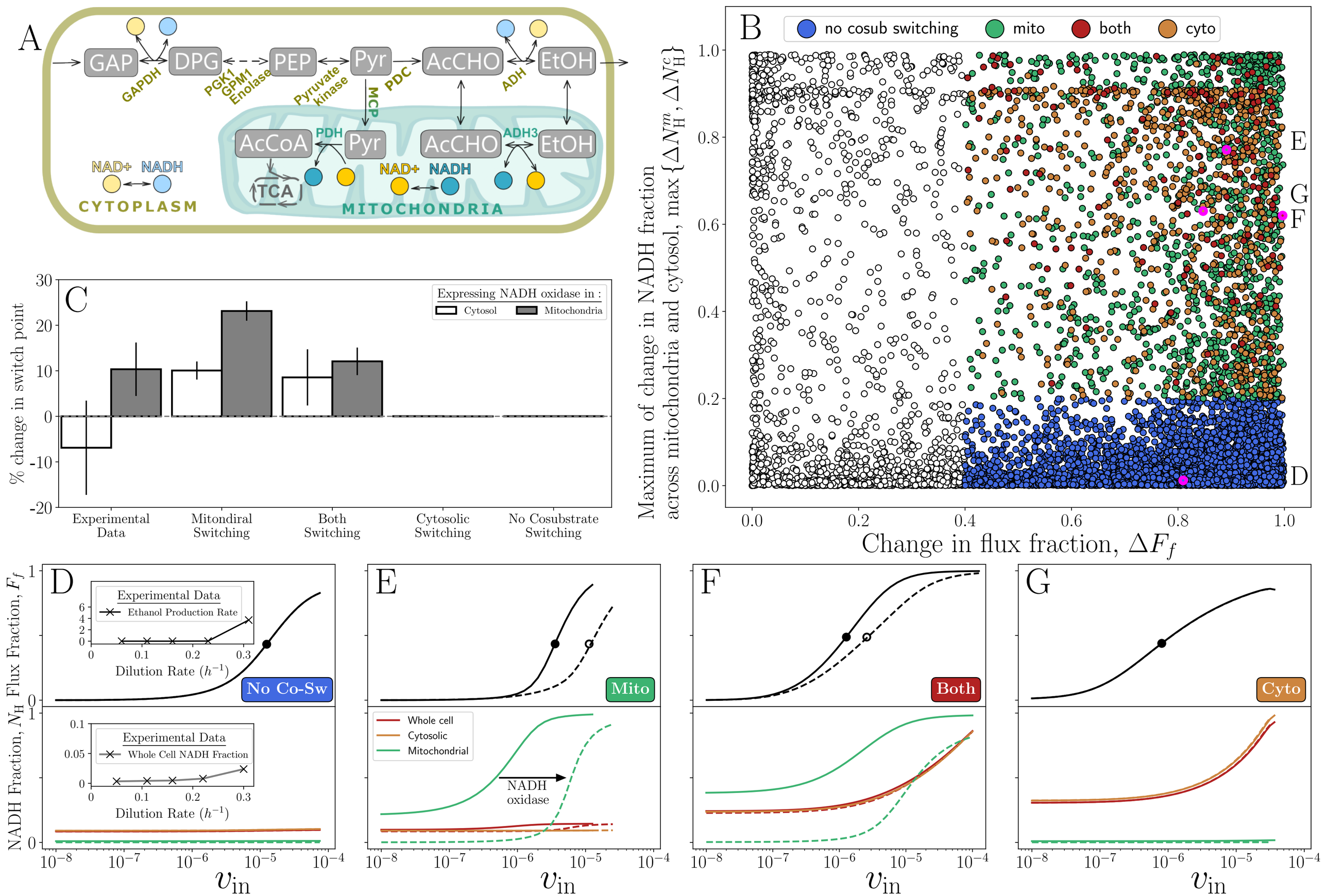
A
GAP
DPG
PEP
Pyr
AcCHO
EtOH
GAPDH
PGK1
GPM1
Enolase
Pyruvate kinase
MCP
PDC
ADH
AcCoA
PDH
Pyr
AcCHO
ADH3
EtOH
NAD+
NADH
TCA
CYTOPLASM
MITOCHONDRIA
B
no cosub switching
mito
both
cyto
Maximum of change in NADH fraction across mitochondria and cytosol, $\max\{\Delta N_H^m, \Delta N_H^c\}$
Change in flux fraction, ΔF_f
C
Expressing NADH oxidase in :
Cytosol
Mitochondria
% change in switch point
Experimental Data
Mitondiral Switching
Both Switching
Cytosolic Switching
No Cosubstrate Switching
D
E
F
G
Experimental Data
Ethanol Production Rate
Dilution Rate (h^{-1})
Whole Cell NADH Fraction
No Co-Sw
Mito
Both
Cyto
Whole cell
Cytosolic
Mitochondrial
NADH oxidase
Flux Fraction, F_f
NADH Fraction, N_H
v_{in}